Proposed Solution

iMovies MERN Project • Phase-wise Documentation

Project: iMovies MERN **Document:** Proposed Solution

1. Executive summary

iMovies is a full-stack web application intended to simplify movie discovery and ticket-booking workflows. It uses React for the interface, Express and Node.js for APIs, and MongoDB for persistence.

2. User workflow

- User opens the application and browses or searches movies.
- User registers or signs in.
- User chooses a movie, show date/time and seats.
- The frontend calculates and displays an estimated total.
- Backend validates the request, checks the movie and reserves the requested seats according to its reservation logic.
- User can view the booking in My Bookings and cancel an eligible booking.
- Administrator signs in separately to manage movies and review bookings.

3. Main modules

Module	Responsibilities
Authentication	Registration, login, token/session handling and protected routes.
Movie catalogue	Read movie records and support search/display.
Booking	Validate movie/seats/show data, create bookings and manage cancellation.
Reservation control	Prevent conflicting seat reservations on the server.
Admin	Protected movie management and booking review.
Database	Persist users, movies, bookings and seat reservations.

4. Technology

Frontend: React, Vite and CSS. Backend: Node.js and Express. Database: MongoDB with Mongoose. Authentication: JWT and password hashing with bcrypt. Development tools: VS Code, CMD, npm and Git/GitHub.

5. Security and quality

- Keep credentials and JWT secrets in environment variables, not source control.
- Hash passwords; never store plaintext passwords.
- Validate all input on the server.
- Enforce ownership when a user reads or cancels a booking.
- Protect administrator APIs with role checks.
- Use HTTPS and secure production cookie/token practices when deployed.

- Use clear error handling and avoid returning sensitive server details.

6. Limitations and future enhancements

- Integrate a legitimate payment gateway before accepting real payments.
- Provide server-driven seat availability refresh and concurrency tests.
- Add email/SMS confirmations only with suitable consent and provider setup.
- Add pagination, accessibility checks and responsive-device testing.
- Add logging, monitoring, backups and deployment automation.

7. Expected outcome

A working prototype demonstrating the complete core workflow locally, with clear separation of frontend, backend and database responsibilities.

Prepared for academic project documentation. Verify implementation-specific details and fill in actual test evidence before final submission.